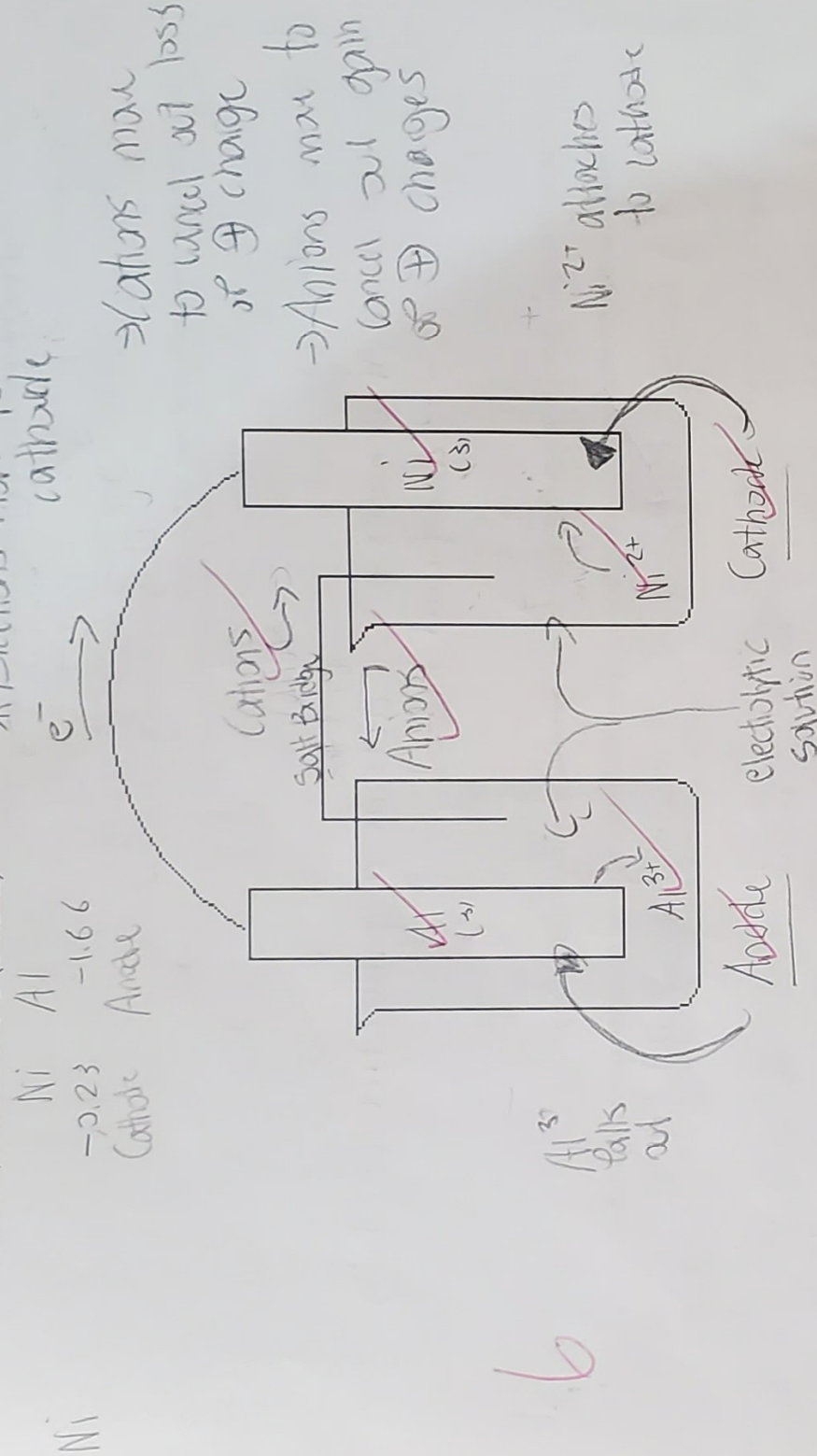


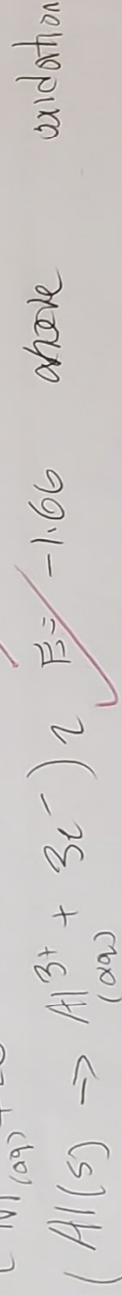
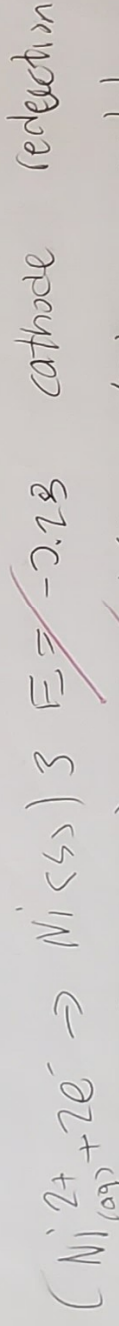
PART C: COMMUNICATION (10 MARKS)

1. A spontaneous reaction occurs in a nickel-aluminum electrochemical cell.

- a) Using the blank galvanic cell below, identify the anode, cathode, electrolyte solutions, salt bridge, flow of electrons, and flow of ions. (6 marks)

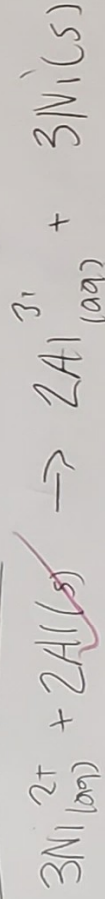


- b) Write the half reactions and calculate the standard cell potential. Make sure to show the overall equation. (4 MARKS)



$$E = -0.23 - (-1.66) = 1.43 \text{ V}$$

Overall equation:



PART D: APPLICATION (6 MARKS)

1. A homeowner is building a deck on the side of a house. She uses galvanized nails for most of the deck but runs out and completes the deck using untreated iron nails. After a few months the ungalvanized nails begin to rust, except for the ones hammered through the aluminum siding which attaches the deck to the house.

- (a) List the minimum requirements for the corrosion of iron. (1 MARK)

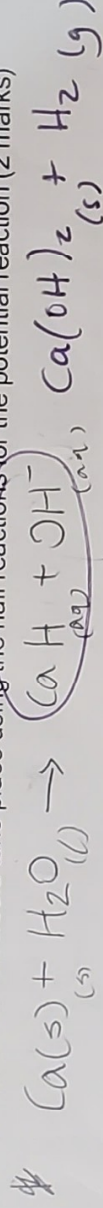
O.S Presence of water in contact with both rusted and uncorroded
Iron. water must take rusted and uncorroded region.

- (b) Explain the observations above. (2 MARKS)

The nails that went through the aluminum siding are protected
! as no water can come into contact with the iron preventing the
corrosion. Al donates e^- - acts as a protective coating
- Sacrificial metal.

2. A piece of calcium metal is accidentally dropped into a sink full of water in a laboratory.

- (a) Predict whether a reaction would take place using the half reactions for the potential reaction (2 marks)



Anode $\text{Ca(s)} \rightarrow \text{CaH} + e^-$ $V = -2.76$
Cathode $2\text{H}_2\text{O} \rightarrow 2\text{OH}^- + \text{H}_2 \text{ (g)}$ $+ 2e^-$ $V = 0.83$
These up
100 k in the table
will occur in the
forward direction
copy directly

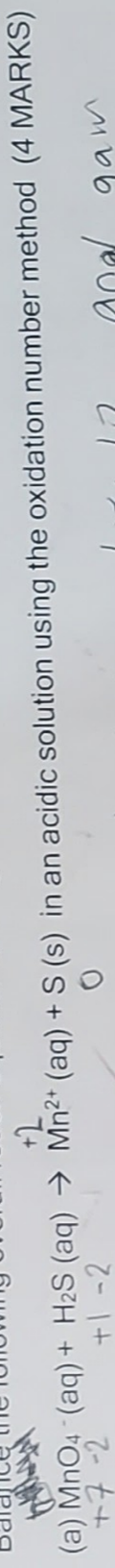
$$E_1 = -0.83 - (-2.76) = 1.93 \text{ V}$$

- (b) What would you expect to observe in this situation? What test could you perform to confirm your prediction? (1 mark)

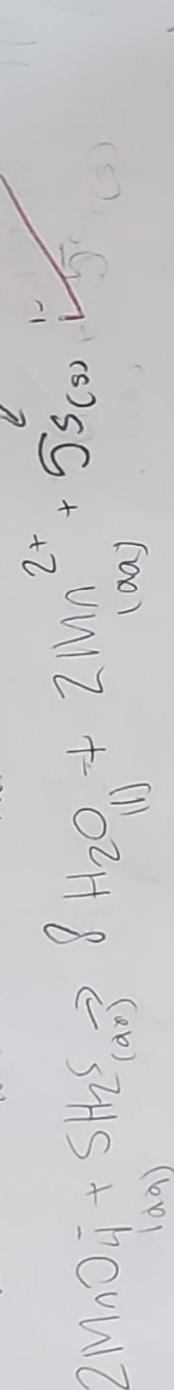
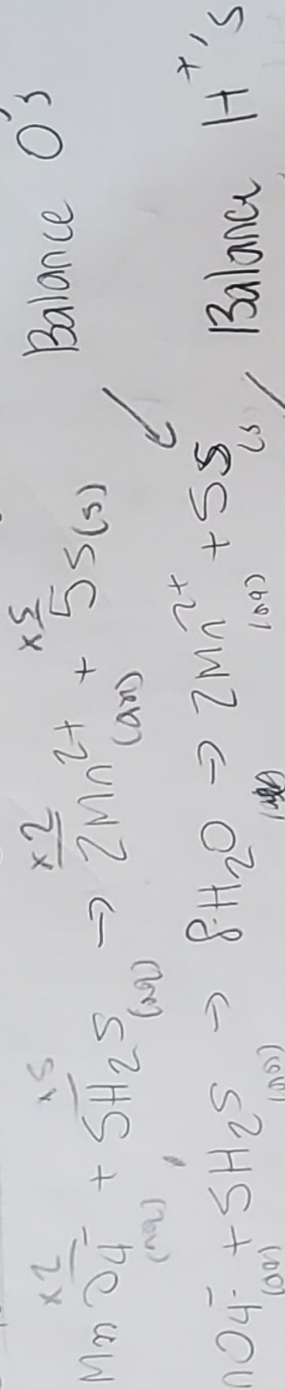
I would expect to observe the metal slowly dissolving
while releasing H_2 gas. To test and confirm
! my prediction I would collect the gas in a
tube and place a burning splint inside. If it
pops then it was the gas, confirming my theoretical
reaction.

PART B: THINKING (21 MARKS)

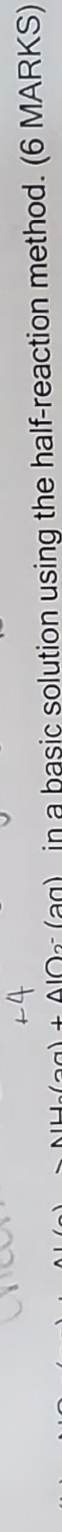
Balance the following overall redox equations, using the method indicated:



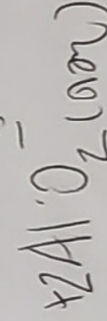
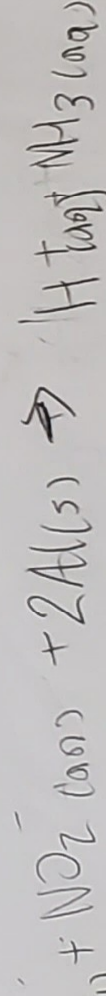
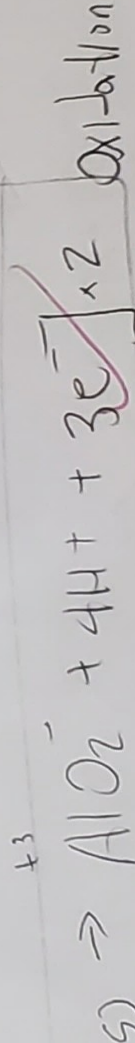
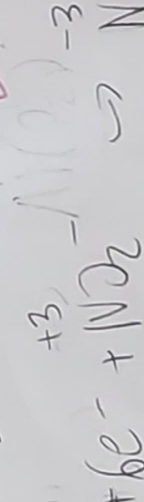
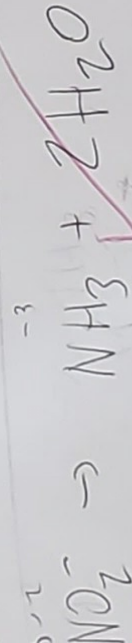
Mn gains 5 electrons $\times 2$ Must both lose 10 and gain 10
S loses 2 electrons $\times 5$ then gained $\bar{e} = \text{lost } \bar{e}$ $2 \times 5 = 5 \times 2$



check out as change of both sides is equal to +4



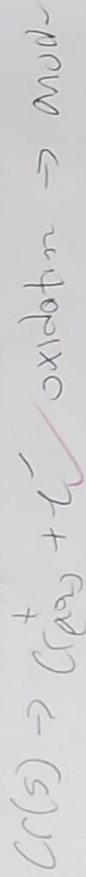
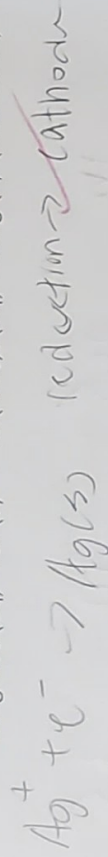
note at both half reactions and balance



the basic we must add OH^- to both sides so then is
 $\text{NO}_2^- + \text{H}_2\text{O} + 2\text{Al} + 2\text{OH}^- \rightarrow \text{NH}_3 + 2\text{AlO}_2^-$

Both same amount of $\bar{e}^-$'s lost/gained

2. Calculate the standard cell potential for the galvanic cell in which the following reaction occurs. Determine whether it will occur spontaneously. SHOW YOUR WORK.



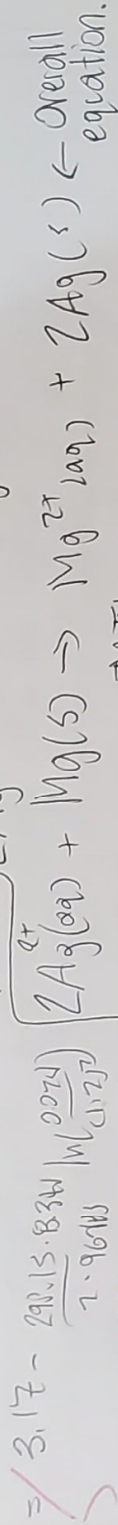
$$E^\circ = E_{\text{cathode}} - E_{\text{anode}} = 0.8 - (-0.73) = 1.53 \text{ V}$$

$E^\circ = 1.53 \text{ V}$
Spontaneous as $E^\circ > 0$

3. Consider the cell: $\text{Mg} | \text{Mg}^{2+} (0.024 \text{ M}) || \text{Ag}^+ (1.2 \text{ M}) | \text{Ag}$. Calculate the cell potential at 25°C. (4 MARKS)

$$E = E^\circ - \frac{RT}{nF} \ln(Q) \quad \text{Mg}(\text{s}) - \text{Mg}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Ag}(\text{s}) \quad E^\circ = -2.37$$

$$= E^\circ - \frac{(298.15)(8.314)}{(2)(96485)} \ln\left(\frac{(0.024)}{(1.2)^2}\right) \quad 2\text{Ag}^{+} + 2e^- \rightarrow 2\text{Ag}(\text{s}) \quad E^\circ = 0.8$$



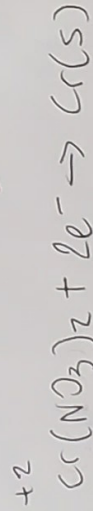
$$E = 3.22 \text{ V}$$

$Q = \frac{[\text{Mg}^{2+}]}{[\text{Ag}^+]^2}$ $n = 2$ $R = 8.314$ $F = 96485$ $T = 25 = 298.15$

4. In an electrolytic cell, a current of 15.0 amps is applied for 4355 seconds to a solution of $\text{Cr}(\text{NO}_3)_3$. What mass of chromium metal is plated out of the solution? (4 MARKS)

Finding how much $Q = 15.4355 = 65325 \text{ C}$

$$\frac{Q}{F} = n \quad \frac{65325}{96485} = 0.677774 \text{ mols of } e^-$$



Every 3 mols of e^- , one mol of $\text{Cr}(\text{s})$

$$\frac{0.677774 \text{ mol } e^-}{3} = \text{mol Cr}$$

$$0.3385 \text{ mol} = \text{Cr}(\text{s})$$

To find mass in grams

$$52 = \text{molar mass g/mol}$$

$$52 \text{ g/mol} \cdot 0.3385 \text{ mol} = 17.6 \text{ grams of Cr is plated out of solution.}$$